

Allamaprabhu Ani

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SUMMARY

Computational mechanics PhD candidate who leads end-to-end development of **PhAST**, an open, matrix-free, differentiable PyTorch solver for phase-field fracture. Builds GPU finite-element software, verified inverse-analysis workflows, and reproducible HPC benchmarks. Experience translating mechanics research into documented software, technical publications, and aerospace analysis automation.

CORE COMPETENCIES

Scientific ML: Python, PyTorch, autograd, CUDA, NumPy/SciPy, scikit-learn, neural operators, PINNs, differentiable simulation, inverse problems.

Numerical engineering: finite elements, phase-field fracture, sparse linear algebra, Akantu, FEniCSx, COMSOL, MSC Nastran/Patran, Abaqus, MATLAB.

Technical delivery: verification and validation, benchmark design, technical documentation, open-source maintenance, Slurm, Linux, Git, GitHub Actions.

EXPERIENCE

PhD Student, Computational Mechanics and ML

Jul 2023 - present (writing-up status)

City St George's, University of London

- Own the architecture, implementation, documentation, and maintenance of **PhAST**, covering matrix-free quasi-static and dynamic phase-field fracture workflows in PyTorch.
- Designed and implemented autograd-compatible fracture-toughness recovery; verified gradients against finite differences before applying the workflow to inverse analysis.
- Established a reproducible evidence suite comprising six benchmark configurations and an approximately one-million-node NVIDIA A100 study, with declared reference comparisons and numerical settings.
- Built reproducible Slurm workflows on Hyperion2 and maintained matching development paths for local Mac CPU and WSL CUDA environments.
- Co-organised two international Mechanics and Machine Learning workshops; built Qualtrics registration and event communications, supported invited speakers, and ran hybrid broadcasting for more than 70 researchers in 2025.

Visiting Researcher, Computational Solid Mechanics

2025

LSMS Lab, EPFL (host: Prof. J.-F. Molinari)

- Led the technical synthesis and manuscript delivery of the 2026 *Engineering Fracture Mechanics* review of ML for computational fracture, coordinating contributions across City, EPFL, and the University of Florida.

Founder and Tool Developer

2020 - 2021

Aeroknacks India Ltd

- Founded the venture and delivered Excel-VBA aerospace analysis tools to an engineering firm, translating E.F. Bruhn, Michael Niu, Boeing Design Manuals, and ESDU methods into repeatable calculation workflows.
- Implemented Cozzone plastic bending, lug strength, fastener load transfer, buckling, modal, and composite ABD tools with documented engineering inputs.

EDUCATION

PhD, Mechanical Engineering, City St George's, University of London (Jul 2023 - present; writing-up status). Thesis: matrix-free differentiable fracture mechanics and inverse analysis in PyTorch.

MTech, Design and Manufacturing, National Institute of Technology Silchar (2021 - 2023). CGPA 9.88/10. Best Student Award.

BE, Mechanical Engineering, GM Institute of Technology, VTU (2016 - 2020). CGPA 8.47/10.

SELECTED PUBLICATIONS

Ani, A.S., Molinari, J.-F., Subhash, G., Ponnusami, S.A. (2026). A matrix-free, differentiable PyTorch solver for phase-field fracture: Formulation, benchmarks, and inverse analysis. [arXiv:2606.23458](https://arxiv.org/abs/2606.23458).

Ani, A.S., Nakka, R., Subhash, G., Molinari, J.-F., Ponnusami, S.A. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics* 332, 111778.

AWARDS

Yeoman and Travelling Scholarship (2026); Dean's Award for Outstanding Teaching Support (2026); AFHEA (2024); Funded PhD Studentship (2023); Best Student Award (2023); Royal Aeronautical Society Member (2026).